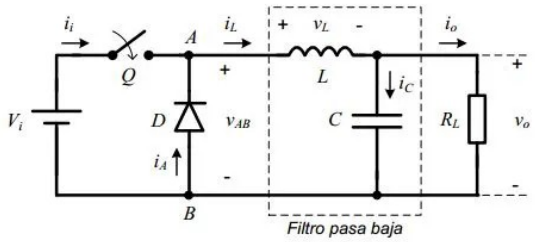
DOC-ANALOG-002

1. **Formulation of the Physical Model of the Plant with Parasites**

To obtain a numerically exact small-signal transfer function from control input to output $G_{vd}(s)$ it is essential to abandon the assumptions of ideal components. In high-precision analog systems, the DC resistance of the inductor (DCR) and the equivalent series resistance of the output capacitor (ESR) significantly affect the damping factor, DC gain, and the location of the cutoff frequencies.



Nominal System Parameters (CCM Operating Point)

Parasitic Element / Parameter $I_{out.max} = 2.0\ A \Rightarrow R_L = \frac{5.0\ V}{2.0\ A}$	**Symbol**	**Physical Value**	**Origin / Physical Criteria**
Input Voltage	V_{in}	24.0 V	Nominal Power Bus
Load Resistance	R_L	2.5 Ω	
Output Inductance	L	68.0 µH	Rated power inductance
DC Resistance of Inductance	R_L *(DCR)*	25.0 mΩ	Losses due to copper wires in the winding
Output Capacity	C_{out}	220.0 µF	Output Bank Capacity
Equivalent Series Resistance	R_C *(ESR)*	25.0 mΩ	Losses in the dielectric/terminals of C_{out}

2. **Formal Derivation of the Transfer Function $G_{vd}(s)$**

Using the state-space averaging method in Continuous Conduction Mode (CCM), the switching node is replaced by an equivalent voltage source with a small-signal average value $v_{sw}(s) = V_{in} \cdot \hat{d}(s)$.

Step 1: Impedance of the Parasitic Capacitive Block $Z_C(s)$

The impedance of the output capacitor branch with its ESR in series is:

$$Z_C(s) = r_C + \frac{1}{sC_{out}} = \frac{1 + sr_C C_{out}}{sC_{out}}$$

Step 2: Equivalent Load Impedance $Z_P(s)$

By connecting the capacitive branch $Z_C(s)$ in parallel with the load resistor R_L:

$$Z_P(s) = Z_C(s)||R_L = \frac{R_L \cdot \left(\frac{1+sr_C C_{out}}{sC_{out}}\right)}{R_L + \frac{1+sr_C C_{out}}{sC_{out}}} = \frac{R_L(1+sr_C C_{out})}{1+sC_{out}(R_L+r_C)}$$

Step 3: Voltage Divider Used in the Plant

The output voltage $\hat{v}_{out}(s)$ is obtained using the voltage divider between $Z_P(s)$ and the series inductive branch ($sL + r_L$):

$$G_{vd}(s) = \frac{\hat{v}_{out}(s)}{\hat{d}(s)} = V_{in} \cdot \frac{Z_P(s)}{Z_P(s) + sL + r_L}$$

Substituting the expression for $Z_P(s)$:

$$G_{vd}(s) = V_{in} \cdot \frac{\frac{R_L(1+sr_CC_{out})}{1+sC_{out}(R_L+r_C)}}{\frac{R_L(1+sr_CC_{out})}{1+sC_{out}(R_L+r_C)} + sL + r_L}$$

By multiplying the numerator and denominator by $[1 + sC_{out} (R_L + r_C)]$:

$$N(s) = V_{in} \cdot R_L(1 + sr_CC_{out})$$

$$D(s) = R_L(1 + sr_CC_{out}) + (sL + r_C)[1 + sC_{out}(R_L + r_C)]$$

Expanding the terms in the denominator $D(s)$:

$$D(s) = R_L + sR_Lr_CC_{out} + sL + r_C + s^2LC_{out}(R_L + r_C) + sr_CC_{out}(R_L + r_C)$$

Grouping terms by powers of the complex variable s:

$$D(s) = (R_L + r_L) + s[L + C_{out}(R_Lr_C + r_LR_L + r_Lr_C)] + s^2[LC_{out}(R_L + r_C)]$$

Factoring ($R_L + r_L$) to obtain the normalized canonical polynomial form:

$$G_{vd}(s) = \left(V_{in}\frac{R_L}{R_L + r_L}\right) \cdot \frac{1 + sr_CC_{out}}{1 + s\left[\frac{L+C_{out}(R_Lr_C+r_LR_L+r_Lr_C)}{R_L+r_L}\right] + s^2\left[\frac{LC_{out}(R_L+r_C)}{R_L+r_L}\right]}$$

3. Second-Order Canonical Form and Numerical Analysis

Expressing $G_{vd}(s)$ in its standard canonical representation in control theory:

$$G_{vd}(s) = G_{vd0} \cdot \frac{1 + \frac{s}{w_z}}{1 + \frac{s}{Qw_0} + \frac{s^2}{w_0^2}}$$

3.1. Direct Current Gain $G_{vd0}(s)$

Due to the voltage drop attenuated by the resistive divider consisting of the *DCR* (r_L) and the load (R_L):

$$G_{vd0}(s) = V_{in} \cdot \frac{R_L}{R_L + r_L} = 24.0\ V \cdot \frac{2.5\ \Omega}{2.5\ \Omega + 0.025\ \Omega} = 24.0\ V \cdot \frac{2.5}{2.525} \approx 23.762\ V$$

On a logarithmic magnitude scale:

$$|G_{vd0}|_{dB} = 20log_{10}(23.762) \approx 27.518\ dB$$

3.2. Zero Introduced by the Capacitor's ESR (w_z, f_z)

The output capacitor's *ESR* (r_C) introduces a real zero in the left half-plane that stops the gain roll-off at high frequencies and injects a phase lead of +90º:

$$w_z = \frac{1}{r_C \cdot C_{out}} = \frac{1}{0.025\ \Omega \cdot 220 \times 10^{-6}} = \frac{1}{5.5 \times 10^{-6}} \approx 181,818.18\ rad/s$$

$$f_z = \frac{w_z}{2\pi} = \frac{181,818.18}{2\pi} \approx 28,952.42\ Hz\ (\approx\ 28.95\ kHz)$$

3.3. Resonant Frequency of the Double Pole (w_0, f_0)

The quadratic term in the denominator defines the natural resonance frequency of the damped LC filter:

$$w_0 = \sqrt{\frac{R_L + r_L}{L \cdot C_{out} \cdot (R_L + r_C)}}$$

Substituting the values from the system:

- Numerator: $R_L + r_L = 2.5 + 0.025 =$ ***2.525 Ω***
- Denominator: $LC_{out}(R_L + r_C) = (68 \times 10^{-6}) \cdot (220 \times 10^{-6}) \cdot (2.5 + 0.025) = 1.496 \times 10^{-8} \cdot 2.525 =$ ***$3.7774 \times 10^{-8}\ s^2 \cdot \Omega$***

$$w_0 = \sqrt{\frac{2.525}{3.7774 \times 10^{-8}}} = \sqrt{66,844,919.785} \approx 8,175.87\ rad/s$$

Calculating the fundamental frequency in Hertz (*Hz*):

$$f_0 = \frac{w_0}{2\pi} = \frac{8,175.87}{2\pi} \approx 1,301.23\ Hz\ (\approx\ 1.301\ kHz)$$

3.4. Resonant Quality Factor (*Q*) and Damping (ζ)

The coefficient of the linear term *s* in the denominator satisfies the relation $1 / Qw_0 = b_1$:

$$b_1 = \frac{L + C_{out}(R_L r_C + r_L R_L + r_L r_C)}{R_L + r_L}$$

Evaluating the combined parasitic terms:

- $R_L r_C = 2.5 \cdot 0.025 = 0.0625\ \Omega^2$
- $r_L R_L = 0.025 \cdot 2.5 = 0.0625\ \Omega^2$
- $r_L r_C = 0.025 \cdot 0.025 = 0.000625\ \Omega^2$
- *Sum of Terms* $R_{comb} = 0.125625\ \Omega^2$

Numerator of b_1:

$$Num_{b1} = 68 \times 10^{-6} + (220 \times 10^{-6} \cdot 0.125625) = 68 \times 10^{-6} + 27.6375 \times 10^{-6} = 95.6375 \times 10^{-6}\ s$$

Dividing by (R_L + r_L) = 2.525 Ω:

$$b_1 = \frac{95.6375 \times 10^{-6}}{2.525} \approx 37.8762 \times 10^{-6}\ s$$

Therefore, the Quality Factor *Q* is:

$$Q = \frac{1}{b_1 \cdot w_0} = \frac{1}{(37.8762 \times 10^{-6} s) \cdot (8,175.87\ rad/s)} = \frac{1}{0.309673} \approx 3.230$$

The dimensionless damping factor ζ is given by:

$$\zeta = \frac{1}{2Q} = \frac{1}{2 \cdot 3.230} \approx 0.1548$$

Interpretation of Resonant Overdrive (Peaking)

Since $Q = 3.230 > 0.707$, the system exhibits severe underdamping near $f_0 = 1.301$ *kHz*. The peak magnitude relative to the DC gain is:

$$ResonantPeak\ (dB) \approx 20log_{10}(Q) = 20log_{10}(3.230) \approx 10.18\ db$$

4. Summary Table of Plant Dynamics (Open-Loop)

$$G_{vd}(s) = 23.762 \cdot \frac{1 + \frac{s}{181{,}818.18}}{1 + \frac{s}{26{,}408.06} + \frac{s^2}{66{,}844{,}919.78}}$$

Dynamic Parameter	Notation	Calculated Exact Value	Physical and Circuit Analysis
Gain in DC	G_{vd0}	23.762 V (27.52 dB)	Mitigated by *DCR* (r_L = 25 mΩ)
LC Resonant Frequency	f_0	1,31.23 Hz (1.301 kHz)	Location of the filter's dual pole
Magnitude Resonance Peak	$\|G_{vd}(f_0)\|_{max}$	+37.70 dB (+10.18 dB vs DC)	Characteristic underdamping (ζ = 0.155)
Resonant Phase Drop	$\angle G_{vd}(f_0)$	-90º (abrupt transition to -180º)	Requires compensation by adv. network
Zero per *ESR* of C_{out}	f_z	28,952.42 Hz (28.95 kHz)	Flatten the slope of -40 to -20 dB/dec

5. Physical Implications for the Design of Type II/III Compensators

- *Severe Phase Delay:* Between 1.0 kHz and 2.0 kHz, the phase collapses from 0º to nearly -180º. If the desired crossover frequency (f_c) is set to 10 kHz ($f_{sw} / 10$), the analog compensator must provide a phase boost of approximately +70º to +80º using zeros located below f_c.
- *Impact of the ESR (r_C):* The zero at $f_z \approx$ 28.95 kHz helps restore +90º of phase above 29 kHz. However, in modern polymer capacitors with *ESR* < 10 mΩ, this zero shifts to much higher frequencies (>75 kHz), forcing the op-amp-based compensator to autonomously provide all compensation using a Type III topology (two zeros and two poles).

6. References and Sources of Contrast

For analytical verification, formal derivation of the state space, and comparison of the models with parasitic impedances (*DCR* and *ESR*) developed in this document, the following sources of technical reference literature are cited:

1. *Middlebrook, R. D., & Ćuk, S. (1976).* A General Unified Approach to Modeling Switching-Converter Power Stages. IEEE Power Electronics Specialists Conference (PESC), 18–34. (A seminal work on the state-space averaging technique used in Section 2 to linearize the plant for small-signal analysis.)

2. *Vorperian, V. (1990).* Simplified analysis of PWM converters using a model of the PWM switch. Part I: Continuous conduction mode. IEEE Transactions on Aerospace and Electronic Systems, 26(3), 490–496.
(Theoretical basis for the equivalent averaged switch model in CCM used in open-loop analytical simulations).

3. *Erickson, R. W., & Maksimović, D. (2001).* Fundamentals of Power Electronics (2nd ed.). Springer Science & Business Media.
(Chapter 7: Small-Signal AC Modeling; Chapter 8: Converter Transfer Functions. Direct source for the canonical derivation of $G_{vd}(s)$, damping ζ, and the calculation of the Q factor).

4. *Basso, C. (2016).* Linear Circuit Transfer Functions: An Introduction to Fast Analytical Techniques (1st ed.). Wiley-IEEE Press.
(Section on the Fast Analytical Techniques [FACTs] method applied to second-order *LC* filters with parasitic resistances r_L and r_C, used to determine the zero f_z and the operating point offset G_{vd0}).

5. *Ridley, R. (2012).* Switching Power Supply Design: Control Loop Cookbook. Ridley Engineering Inc.
(Chapter 2: Small-Signal Modeling of Buck Converters. Comparison Guide for the Impact of Capacitive *ESR* on the Phase Margin Before Compensation).

6. *Texas Instruments / Unitrode. (1999).* SLUP113: Modeling, Analysis, and Compensation of DC/DC Converters (Application Note by Karl H. Kruse). Texas Instruments Literature.
(Mathematical demonstration of phase collapse at -180º at the f_0 resonance and formulation of the frequency response of the power stage).

7. *Analog Devices. (2008).* AN-1138: High-Speed Op-Amp Compensation and Small-Signal AC Loop Stability. Analog Devices Application Note.
(Analysis methodology for calculating pole interaction between the power stage and the input of the error op-amp).